Rohan Malhotra

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EDUCATION

New York University, Courant Institute of Mathematical Sciences — New York, NY
B.A. in Mathematics and Computer Science (Double Major) — *Expected May 2027*

- **GPA:** 3.7/4.0 | Accelerated three-year degree | **Graduate course:** Mathematical Techniques for CS Applications
- **Coursework:** Probability & Statistics, Machine Learning, Linear Algebra, Numerical Methods, Algorithms

EXPERIENCE

IBM — Dallas, TX
Applications Developer Intern, IBM Consulting — *May 2026 – Aug 2026*

- Built an XGBoost cash-flow model and 13-day rolling cash forecasts in Oracle EPM Planning for finance teams.
- Built SOFR loan, interest-accrual, and close-report models; piped banking API data into ERP and EPM as JSON.
- Fine-tuned Qwen-Coder-32B on Oracle EPM assets, raising EPM build-generation accuracy from **36.7% to 95.0%**.

Robotics Engineer Intern — *Jun 2026 – Aug 2026*

- Labeled 898 images and trained YOLO11 models to 99.5% mAP@50 with 99.4% precision and 99.4% recall.

Aress Software — New York, NY
Machine Learning Intern — *Jun 2025 – Aug 2025*

- Built Power BI dashboards surfacing 20+ real-time KPIs for global operations teams; managed the JIRA workflow.
- Designed NoSQL pipelines unifying 10+ data sources into one analytics layer; optimized SQL queries for speed.
- Trained and deployed PyTorch models on **500K+ multimodal records** across Azure and AWS.

Y-Axis Overseas Careers — Chatham, NJ
Data Engineering Intern — *May 2024 – Aug 2024*

- Cleaned and standardized 20K+ SQL records with schema fixes and validation, cutting data-prep time by **15%**.
- Engineered 12 predictive features from dataset analysis and transformations, raising model accuracy by **10%**.
- Built automated Excel dashboards with dynamic filtering, enabling faster eligibility reviews.

Kalshi — New York, NY
Quantitative Developer — *Jan 2026 – May 2026*

- Modeled job-loss risk as a Poisson hazard on FRED and BLS data; measured hedge tail-risk cuts via Monte Carlo.
- Built a Next.js/Python engine that recommends hedge size and compares cost vs. tail risk using FRED and BLS data.

RESEARCH

Reddit Data in Quantitative Financial Models (published) — Jan 2025 – May 2025
Co-author, Virginia Tech (VTechWorks) — *hdl.handle.net/10919/124730*

- Co-authored a published paper on how r/wallstreetbets sentiment after the GameStop squeeze relates to volatility.
- Applied NLP sentiment scoring and regression models in Python to quantify sentiment's impact on equity returns.

PROJECTS

Love Island Sponsorship ROI Analysis | *Python, Pandas, Plotly* — 2025

- Compared sponsorship cost, reach, and engagement across Love Island brand deals; built an interactive ROI page.

SPY Gradient Boosting Predictor | *scikit-learn, PCA, yfinance* — 2025

- Trained a gradient-boosting model to predict the next SPY close from EMA, RSI, and MACD features with PCA.

Refrax | refrax.io | *C++, Next.js, TypeScript, PostgreSQL* — Jun 2025 – Feb 2026

- Built a quantitative research and backtesting platform with 2D/3D visualization of financial data.
- Designed low-latency REST/WebSocket pipelines streaming market data into a PostgreSQL research environment.

SKILLS

Languages: Python, SQL, R, C++, MATLAB, Java
Data & ML: Pandas, NumPy, SciPy, statsmodels, scikit-learn, XGBoost, PyTorch, NLP, regression, Monte Carlo
Tools: Power BI, Excel (macros), PostgreSQL, NoSQL, Jupyter, Plotly, AWS, Azure, Docker, Git, Oracle EPM